Mohammed FELLAJI

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Education

Ecole CentraleSupélec - Loria lab — Metz, France
PhD on Bayesian Deep Learning — *01/2022 – 07/2025*

- Supervisors: Miguel Couceiro and Frédéric Pennerath
- Title: "Epistemic Calibration for Bayesian Deep Learning: Principles, Issues and Solutions"

Ecole CentraleSupélec — Metz, France
Engineering School - Major: Data Science — *09/2017 – 04/2021*

University of Lorraine — Metz, France
Master of Mathematics (M2): fundamental and applied mathematics — *09/2020 – 04/2021*

Ecole Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
Exchange program — *02/2019 – 07/2019*

Prep School — Tangier, Morocco
Maths-physics Major (Preparatory classes for the top French engineering schools) — *09/2015 – 06/2017*

Experience

Postdoctoral Researcher — Luxembourg, Luxembourg
University of Luxembourg - SnT — *01/2026 – current*

- Working on the AI Assessment Sandbox Configurator: an open-source platform for evaluating and assessing risks in AI models (https://github.com/lux-ai-factory/aisc)

Teacher assistant — Metz, France
CentraleSupélec/IUT Metz — *09/2022 – 04/2024*

- Statistical Models, Algorithms for Data Science
- Introduction to algorithms (VBA), Introduction to Python

Deep Learning Research Intern — Lannion, France
Orange Labs — *05/2021 – 10/2021*

- **Keywords:** Audio Processing, Audio Embeddings, Large Pre-trained Models, Fine-tuning, Deep Learning.
- **Researched and applied** deep learning techniques, including large pre-trained models and fine-tuning, to generate *audio embeddings* from extensive datasets.
- **Analyzed and compared** novel audio embedding approaches against traditional methods, demonstrating superior performance and efficiency.
- **Demonstrated significant results** by testing audio embeddings on DCASE datasets, enabling the use of smaller models for classification and regression tasks.

Data Science Intern — Berlin, Germany
AiSight GmbH — *03/2020 – 08/2020*

- **Keywords:** IoT, Predictive Maintenance, Anomaly Detection, Vibrations Data, Gaussian Processes, Deep Learning.
- **Engineered** classification and regression models using high-frequency IoT sensor data for **predictive maintenance**. Utilized data visualization, cleaning, signal processing, Gaussian processes, and deep learning algorithms, presenting results directly to clients.
- **Developed and optimized anomaly detection** for continuous vibration data; automated client reporting via Slack/email for actionable insights.

Machine Learning Intern Nancy, France
Zhor-Tech *08/2019 – 02/2020*

- **Keywords:** IoT, IMU Sensors, Gait Analysis, Machine Learning, Signal Processing, Data Fusion.
- **Analyzed and processed** IMU sensor data for **gait analysis**, developing multi-class classification models that predicted human movement from 3 distinct classes, enhanced by data augmentation.
- **Developed** a precise penalty kick simulation algorithm in Python using IMU sensor data, presented at *CES 2020* in partnership with **Ubisoft**. Collaborated with the firmware team to develop and test a C version.
- **Optimized C code** for embedded systems, reducing computation error from 20% to 3% and significantly improving memory usage.

Papers

Epistemic Calibration for Bayesian Deep Learning: PhD Thesis
Principles, Issues and Solutions *hal.science/tel-05558187v1*

On the Calibration of Epistemic Uncertainty: ECML PKDD 2024
Principles, Paradoxes and Conflictual Loss *arXiv: 2407.12211*

The Epistemic Uncertainty Hole: CAp 2023
An issue of Bayesian Neural Networks *arXiv: 2407.01985*

Previous Projects

https://github.com/fellajimed/DocSeer *Hugging Face, LangChain, ChromaDB, Faiss, Pydantic ...*

- An intelligent PDF analysis tool that allows you to summarize documents and ask questions about their contents using natural language. It leverages modern language models to provide fast, accurate insights from complex files.

https://github.com/fellajimed/FLAIR *Python, PyTorch, tifffile, Scikit-learn*

- Developed and implemented modular Encoder-Decoder and U-Net models for **image segmentation** on the FLAIR challenge. Utilized data patching for efficient processing of large inputs and to enhance batch diversity.
  The implementation allow streamlining training, logging, and result aggregation.

https://github.com/fellajimed/Conflictual-Loss *Machine Learning, PyTorch, Yaml, NumPy ...*

- Implementation of the published papers. Developed code for training models and aggregating results based in a modular fashion based on a Yaml file.

Skills

Programming Languages: Python, C++, SQL, R, Matlab, Java.

Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn ...

Math: Statistics, probability, calculus, time series, signal processing, regression, optimization, stochastic processes.

About Me

Languages: English (advanced), French (bilingual), Arabic (mother tongue), German (beginner).

Hobbies: Football, Handball, Badminton, Running, Reading, Traveling.

Personality: Autonomy, Innovation, Organization, Team spirit, Project management.

Dev environment: Python, PyTorch, NumPy, Joblib, Git, Vim/NeoVim, Tmux, Linux.

Awards

Recipient of the Moroccan Government Merit Scholarship (2017).

Second place at the maths Olympiad organized by the Spanish school "Ramón Y Cajal" in Tangier (2012).